

Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala, Punjab, India

Department of Electronics and Communication Engineering

M. Tech. (VLSI)

Course Code:-PVL-207

Semester:- II

Course Name:- Low Power System Design

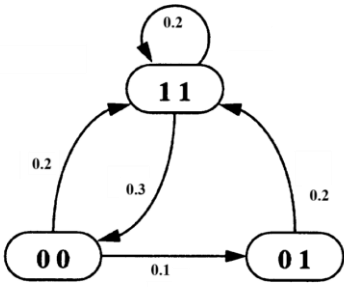
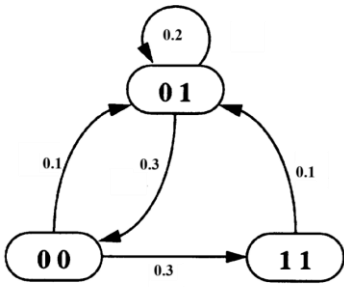
May 21th, 2024; EST

Tuesday, 02:00 P.M. – 05:00 P.M.

Time: 03 hrs., Max. Marks: 40

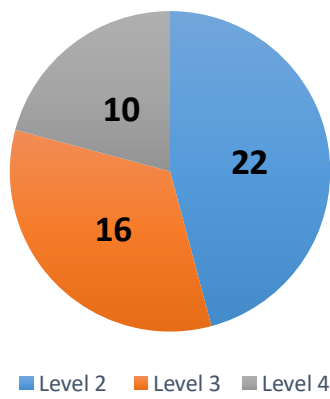
Name of Faculty: Dr. Mohammad Faseehuiddin

Note: Attempt any five questions (Write mathematical expressions to support your answers wherever applicable).

Q. No.	Questions	Marks	CO	BL
Q.1 (a)	Explain the concept of probabilistic power analysis. How accurate it is compared to simulation-based power analysis? Using the relationship between the static probability and frequency find the relation between conditional probability and frequency?	(5)	C04	L2
Q.1 (b)	Compute the transition density and static probability of $y = ab + c$, given $P(a)=0.3$, $P(b)=0.2$, $P(c)=0.5$, $D(a)=1$, $D(b)=3$, $D(c)=2$.	(3)	C04	L4
Q.2(a)	In relation to the low power RAM design discuss in detail two low power design strategies that can be employed in the design of bit lines and word lines. Include the technical details and related diagrams.	(5)	C05	L2
Q.2(b)	Discuss the organization of a random-access memory. Discuss the banking technique in RAM design.	(3)	C05	L3
Q.3(a)	Explain the concept of transistor sizing and pin reordering with the help of examples?	(3)	C03	L3
Q.3(b)	What is a double edge triggered flip flop? How can it help in power reduction explain?	(3)	C03	L2
Q.3(c)	How a self-gating flip-flop employed in low power digital design?	(2)	C03	L2
Q.4(a)	Explain gate reorganization and logic encoding techniques with proper examples? How each technique results in power reduction?	(3)	C03	L3
Q.4(b)	Calculate the state bit transition $E[M]$ for the two state machines. Which of the two machines is more power efficient?   M1 M2	(3)	C04	L4
Q.4(C)	Explain glitch and clock skew?	(2)	C02	L2

Q.5(a)	Explain clock gating and power gating techniques? How clock gating can be implemented explain with a suitable example.	(3)	C03	L3
Q.5(b)	Explain the different types of leakage that occur in digital circuits with proper diagram and mathematical expressions.	(3)	C01	L2
Q.5(c)	What effect does constant field scaling have on (i) doping densities (ii) gate oxide thickness(iii) power density (iv) threshold voltage.	(2)	C02	L2
Q.6 (a)	Discuss the gate capacitance in MOSFET with proper mathematical expressions. How the gate capacitance varies with the bias conditions?	(4)	C03	L3
Q.6 (b)	Explain Variable threshold voltage CMOS (VTCMOS) approach and Dual-Vt assignment approach (DTCMOS) in detail with a design example.	(4)	C03	L4

**Bloom's Level wise Marks
Distribution**



**Course Outcome wise Marks
Distribution**

